

TWO DETAILED COMPUTATIONS OF $\mathrm{THH}(\mathbb{F}_p)$

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ABSTRACT. In the following document, we present two different proofs of Bokstedt periodicity: first, we explain an algebraic argument using group rings, and then we present a more topological argument via Thom spectra. Lastly, we describe a comparison of the two. This document serves as notes for Lecture 6 (Bökstedt periodicity and quotients of DVRs I) of the 2025 Computational Methods in Algebraic K-theory seminar.

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1. INTRODUCTION

Our goal in this write-up is to explain two proofs of the following theorem:

Theorem 1.1 (Bökstedt). We have that

$$\mathrm{THH}_*(\mathbb{F}_p) \cong \mathbb{F}_p[x], \quad |x| = 2.$$

In particular, there is an equivalence of $\mathbb{E}_1$ -algebras:

$$\mathrm{THH}(\mathbb{F}_p) \simeq \mathbb{F}_p[\Omega S^3].$$

The proofs will be of different flavors: one more algebraic and one more topological. The former originates in a paper of Krause and Nikolaus [KN19], and the latter is given in greater generality by [BCS10] (a more detailed discussion of this particular computation found in [KN18]).

In this document, we will use the following definition of THH :

Definition 1.1. Let R be an associative ring spectrum. Then we define

$$\mathrm{THH}(R) = R \otimes_{R \otimes R^{\mathrm{op}}} R.$$

Tensor (smash) products of spectra will be over the sphere spectrum $\mathbb{S}$ unless otherwise noted. Also, we will freely use the language and ideas of higher algebra; we will denote the ∞ -category of spaces by $\mathcal{S}$ and the symmetric monoidal stable ∞ -category of spectra by Sp . Other definitions will be introduced as they become necessary.

2. THE ALGEBRAIC ARGUMENT

In some sense, computing $\mathrm{THH}(\mathbb{F}_p)$ is equivalent to understanding the dual Steenrod algebra $\pi_*(\mathbb{F}_p \otimes \mathbb{F}_p) = H_*(\mathbb{F}_p; \mathbb{F}_p)$, i.e. the lower tensor product in our definition of THH . The first proof we present in these notes reduces to proving the following result:

Theorem 2.1. There is an equivalence of $\mathbb{E}_2$ -ring spectra

$$\mathbb{F}_p \otimes \mathbb{F}_p \simeq \mathbb{F}_p[\Omega^2 S^3].$$

Stated in words, this just means: “The dual Steenrod algebra is the free $\mathbb{E}_2$ - $\mathbb{F}_p$ -algebra on a single generator in degree 1.” We’ll now explain why.

Let A be another $\mathbb{E}_2$ - $\mathbb{F}_p$ -algebra, and let

$$\mathrm{Alg}_{\mathbb{F}_p}^{\mathbb{E}_2} := \mathrm{Alg}_{\mathbb{E}_2}(\mathrm{Mod}_{\mathbb{F}_p})$$

be the category of $\mathbb{E}_2$ - $\mathbb{F}_p$ -algebras. Then we have that:

$$\begin{aligned} \mathrm{Map}_{\mathrm{Alg}_{\mathbb{F}_p}^{\mathbb{E}_2}}(\mathbb{F}_p[\Omega^2 S^3], A) &\simeq \mathrm{Map}_{\mathrm{Alg}_{\mathbb{E}_2}(\mathrm{Sp})}(\Sigma_+^\infty \Omega^2 S^3, A) \\ &\simeq \mathrm{Map}_{\mathrm{Alg}_{\mathbb{E}_2}(\mathcal{S})}(\Omega^2 S^3, \Omega^\infty A) \\ &\simeq \mathrm{Map}_{\mathcal{S}_*}(S^1, \Omega^\infty A) \end{aligned}$$

through a chain of adjunctions. In particular, we pass through the free-forgetful adjunction $\mathrm{Sp} \rightleftharpoons \mathrm{Mod}_{\mathbb{F}_p}$, suspension-loops $\Sigma_+^\infty \dashv \Omega^\infty$, and the free $\mathbb{E}_n$ -space construction $\Omega^n \Sigma^n(-)$. In other words, elements of

$$\pi_0 \mathrm{Map}_{\mathcal{S}_*}(S^1, \Omega^\infty A) \cong \pi_1(A)$$

correspond exactly to homotopy classes of $\mathbb{F}_p$ -linear $\mathbb{E}_2$ maps

$$\mathbb{F}_p[\Omega^2 S^3] \rightarrow A,$$

in a coherent way; i.e. $\mathbb{F}_p[\Omega^2 S^3]$ is universally determined by a single generator in degree 1 as an $\mathbb{E}_2$ - $\mathbb{F}_p$ -algebra.

Assume that we do prove Theorem 2.1, i.e. that $\mathbb{F}_p \otimes \mathbb{F}_p \simeq \mathbb{F}_p[\Omega^2 S^3]$. Then the final computation of THH follows almost immediately, as seen below:

2.1. After this, we're done! Note that $\mathbb{F}_p[-]$ preserves colimits because $\mathbb{F}_p \otimes -$ and $\Sigma_+^\infty(-)$ are both left adjoints. It also sends products to tensor products, since both $\mathbb{F}_p \otimes -$ and $\Sigma_+^\infty(-)$ are symmetric monoidal. Then we have that

$$\begin{aligned}
\mathrm{THH}(\mathbb{F}_p) &= \mathbb{F}_p \otimes_{\mathbb{F}_p \otimes \mathbb{F}_p} \mathbb{F}_p \\
&\simeq \mathbb{F}_p \otimes_{\mathbb{F}_p[\Omega^2 S^3]} \mathbb{F}_p \\
&\simeq \mathbb{F}_p[*] \otimes_{\mathbb{F}_p[\Omega^2 S^3]} \mathbb{F}_p[*] \\
&\simeq \mathrm{colim}(\cdots \rightrightarrows (\mathbb{F}_p[*] \otimes_{\mathbb{F}_p[\Omega^2 S^3]} \mathbb{F}_p[*]) \rightrightarrows (\mathbb{F}_p[*] \otimes_{\mathbb{F}_p[*]}) \\
&\simeq \mathbb{F}_p[\mathrm{colim}(\cdots \rightrightarrows (* \times \Omega^2 S^3 \times *) \rightrightarrows * \times *)] \\
&\simeq \mathbb{F}_p[B\Omega^2 S^3] \\
&\simeq \mathbb{F}_p[\Omega S^3]
\end{aligned}$$

as desired. In the end, we obtain an $\mathbb{E}_1$ -equivalence; all maps preserved multiplicativity except the taking the bar construction, which decreased us to an $\mathbb{E}_1$ equivalence of ring spectra. Here we also used the fact that a model for the classifying space BG is given by the “colimit of a simplicial diagram” bar construction $\mathrm{Bar}(*, G, *)$.

And so $\mathrm{THH}(\mathbb{F}_p)$ is a free $\mathbb{E}_1$ - $\mathbb{F}_p$ -algebra on a generator in degree 2. Thus the question remains: how do we see that the dual Steenrod algebra has the free structure we want?

2.2. Results of Araki-Kudo, Dyer-Lashof, and Steinberger. One might guess that in order to prove that

$$\mathbb{F}_p \otimes \mathbb{F}_p \simeq \mathbb{F}_p \otimes \Sigma_+^\infty \Omega^2 S^3,$$

we ought to look explicitly at the $\mathbb{E}_2$ Dyer-Lashof operations on homology, and use them to show that

$$H_*(\mathbb{F}_p; \mathbb{F}_p) \cong H_*(\Omega^2 S^3; \mathbb{F}_p),$$

especially since these things seem like they would be computable with very classical methods. Indeed, let us recall that for a $\mathbb{E}_2$ -ring spectrum X , we have the Dyer-Lashof operations on $\mathbb{F}_p$ homology:

$$\begin{aligned}
p = 2 : \quad Q^i &: H_k(X; \mathbb{F}_2) \rightarrow H_{k+i}(X; \mathbb{F}_2) \\
p > 2 : \quad Q^i &: H_k(X; \mathbb{F}_p) \rightarrow H_{k+2i(p-1)}(X; \mathbb{F}_p) \\
\beta Q^i &: H_k(X; \mathbb{F}_p) \rightarrow H_{k+2i(p-1)-1}(X; \mathbb{F}_p)
\end{aligned}$$

which satisfy certain conditions, formulas, and relations such that they make sense (e.g. the Cartan formula and the Adem and Nishida relations.)

Then we have the following calculational results:

Lemma 2.1. (Araki-Kudo, Dyer-Lashof, and Steinberger.)

(1) For $p = 2$, we have that

$$H_*(\Omega^2 S^3; \mathbb{F}_2) \cong \mathbb{F}_2[x_1, x_2, \dots]$$

where $|x_i| = 2^i - 1$, and $x_{i+1} = Q^{2^i} Q^{2^{i-1}} \cdots Q^4 Q^2 x_1$; i.e., it is generated by x_1 over the Dyer-Lashof algebra.

(2) For $p > 2$, we have that

$$H_*(\Omega^2 S^3; \mathbb{F}_p) \cong \mathbb{F}_p[y_0, y_1, \dots; z_1, z_2, \dots]/(y_i^2)$$

where $|y_i| = 2p^i - 1$, and $y_{i+1} = Q^{p^i} \cdots Q^p Q x_1$; and $|z_i| = 2p^i - 2$, $z_i = \beta y_{i+1}$; i.e., it is generated by y_0 over the Dyer-Lashof algebra.

(3) The same isomorphisms hold for $\mathcal{A}_* = H_*(\mathbb{F}_p; \mathbb{F}_p)$, though we use different symbols;

$$p = 2 : \quad \mathcal{A}_* \cong \mathbb{F}_2[\xi_1, \xi_2, \dots]$$

$$p > 2 : \quad \mathcal{A}_* \cong \mathbb{F}_p[\tau_0, \tau_1, \dots; \xi_1, \xi_2, \dots]/(\tau_i^2)$$

with the analogous relations; i.e. when $p = 2$, $\mathcal{A}_*$ is generated by ξ_1 over the Dyer-Lashof algebra, and when $p > 2$, $\mathcal{A}_*$ is generated by τ_0 over the Dyer-Lashof algebra.

The first result is [KA56, Theorem 7.1], which computes $H_*(\Omega^N S^n; \mathbb{F}_2)$ for all $N < n$. The second is [DL62, Theorem 5.2], which computes $H_*(\Omega^N S^n; \mathbb{F}_p)$ for all $N < n$ with $p > 2$. The third is [BMMS86, Theorem 2.2, 2.3, 2.4].

The first two are applications of Serre's spectral sequence and transgressions, and the third is a combinatorial and inductive argument based on the Nishida relations and Milnor's original formulation (in terms of dual elements to Steenrod operations).

2.3. Finishing up the algebraic proof. So, now that we've gone through the classical results that we needed, we are effectively done! To be more explicit, let's return to the original problem, i.e. showing that we have the $\mathbb{E}_2$ -equivalence of ring spectra

$$\mathbb{F}_p \otimes \mathbb{F}_p \simeq \mathbb{F}_p[\Omega^2 S^3].$$

Proof of Theorem 2.1. We now know that both $(\mathbb{F}_p)_*(\Omega^2 S^3)$ and $\mathcal{A}_*$ are generated by a single element in degree 1 under the $\mathbb{E}_2$ Dyer-Lashof operations. Then since $(\mathbb{F}_p)_*(\Omega^2 S^3)$ is the free $\mathbb{E}_2$ - $\mathbb{F}_p$ -algebra on a generator in degree 1, this means that the element $\xi_1 \in \pi_1(\mathbb{F}_p \otimes \mathbb{F}_p)$ corresponds to an $\mathbb{E}_2$ map exists

$$f : \mathbb{F}_p[\Omega^2 S^3] \rightarrow \mathbb{F}_p \otimes \mathbb{F}_p$$

determined by the fact that it sends x_1 to ξ_1 in degree 1. This is an isomorphism on homotopy, since $\mathbb{E}_2$ -maps preserve $\mathbb{E}_2$ Dyer-Lashof operations, and therefore an $\mathbb{E}_2$ equivalence. $\square$

3. THE HOPKINS-MAHOWALD RESULT

The following section explains the more topological version of the proof of Theorem 2.1, which rests on a result shown by Mahowald for $p = 2$ and by Hopkins for $p > 2$. A general construction computing THH of Thom spectra is given by [BCS10]. Before we state this result, let's summarize our strategy for this section.

The game-plan is as follows.

- (1) Design a map f_p whose Thom spectrum Mf_p is $\mathbb{F}_p$, at least on $\mathbb{E}_2$ structures.
- (2) Apply a formula for THH of Thom spectra.

3.1. The statement. Now let's actually give the Hopkins-Mahowald result. First, let us consider the following calculation:

$$\pi_1(BGL_1(\mathbb{S}_p^\wedge)) \cong \pi_0(\Omega BGL_1(\mathbb{S}_p^\wedge)) \cong \pi_0(GL_1(\mathbb{S}_p^\wedge)) \cong \pi_0(\mathbb{S}_p^\wedge)^\times \cong \mathbb{Z}_p^\times$$

where we used the fact that $\pi_0(\mathbb{S}_p^\wedge) \cong \mathbb{Z}_p$. Consider the element

$$1 + p \in \mathbb{Z}_p^\times \cong \pi_1(BGL_1(\mathbb{S}_p^\wedge)).$$

Let it be represented by a map $S^1 \rightarrow BGL_1(\mathbb{S}_p^\wedge)$ which induces a two-fold loop map

$$f_p : \Omega^2 S^3 \rightarrow BGL_1(\mathbb{S}_p^\wedge).$$

This map is an $\mathbb{E}_2$ map, since $\Omega^2 \Sigma^2 S^1 \simeq \Omega^2 S^3$ is the free $\mathbb{E}_2$ algebra on S^1 , e.g. we have

$$\mathrm{Map}_{\mathbb{E}_2}(\Omega^2 \Sigma^2 S^1, BGL_1(\mathbb{S}_p^\wedge)) \simeq \mathrm{Map}_{\mathcal{S}_*}(S^1, BGL_1(\mathbb{S}_p^\wedge)),$$

or more generally, $\Omega^n \Sigma^n X$ is the free $\mathbb{E}_n$ algebra on a connected, pointed space X .

Let Mf_p be the Thom spectrum of this map. Since f_p was an $\mathbb{E}_2$ map, Mf_p is also $\mathbb{E}_2$. Then:

Theorem 3.1 (Hopkins-Mahowald). There is an equivalence of $\mathbb{E}_2$ ring spectra $Mf_p \simeq \mathbb{F}_p$.

Since $\Omega^2 S^3 \simeq B\Omega^3 S^3$, we have that $Mf_p \simeq (\mathbb{S}_p^\wedge)_{h\Omega^3 S^3}$. The remainder of this section will be devoted to proving this result.

3.2. The proof. Let's begin by showing that $\pi_0(Mf_p) \cong \mathbb{F}_p$. Then we'll use the 0th Postnikov stage to get our map $Mf_p \rightarrow \mathbb{F}_p$. Specifically, since

$$\pi_0 : \mathrm{Sp}^{\mathrm{cn}} \rightarrow \mathrm{Sp}^\heartsuit \simeq N(\mathrm{Ab})$$

as a functor on connective spectra is left adjoint to the Eilenberg-MacLane spectrum functor H , it preserves colimits. Hence we have that

$$\begin{aligned} \pi_0(Mf_p) &= \pi_0((\mathbb{S}_p^\wedge)_{h\Omega^3 S^3}) \\ &= \pi_0 \left(\mathrm{colim}_{B\Omega^3 S^3} (\mathbb{S}_p^\wedge) \right) \\ &\cong \mathrm{colim}_{B\Omega^3 S^3} (\pi_0(\mathbb{S}_p^\wedge)) \\ &\cong \mathrm{colim}_{B\pi_0 \Omega^3 S^3} (\pi_0(\mathbb{S}_p^\wedge)) \\ &\cong (\pi_0(\mathbb{S}_p^\wedge))_{\pi_0 \Omega^3 S^3} \\ &\cong (\mathbb{Z}_p)_{\mathbb{Z}} \end{aligned}$$

where the $\mathbb{Z}$ action on $\mathbb{Z}_p$ is given by multiplication by $1+p$ by construction. In particular, we can reduce our colimit to being a 1-colimit, because it lands in Ab , which is 1-truncated.

Then we have that

$$\pi_0(Mf_p) \cong (\mathbb{Z}_p)_{\mathbb{Z}} \cong \mathrm{Coeq}(\mathbb{Z}_p \xrightarrow[1]{1+p} \mathbb{Z}_p) \cong \mathbb{Z}_p / (1 - (1+p)) \cong \mathbb{F}_p.$$

Hence, we can now consider the map

$$\varphi : Mf_p \rightarrow \mathbb{F}_p$$

down to the 0th stage of the Postnikov tower of Mf_p . Since the 0th truncation of connective spectra is symmetric monoidal, φ is an $\mathbb{E}_2$ map of ring spectra [Lur17, 2.2.1.10].

Since both Mf_p and $\mathbb{F}_p$ are connective and p -complete (Mf_p is p -complete because it is p -torsion, as the unit of $\pi_*(Mf_p)$ is $1 \in \pi_0(Mf_p) \cong \mathbb{F}_p$) it suffices to show that

$$Mf_p \otimes \mathbb{F}_p \simeq \mathbb{F}_p \otimes \mathbb{F}_p$$

i.e. that φ induces an isomorphism on $\mathbb{F}_p$ -homology

$$\varphi_* : H_*(Mf_p; \mathbb{F}_p) \cong H_*(\mathbb{F}_p; \mathbb{F}_p).$$

Recall that we can “change fiber” for Thom spectra; i.e. if we have a map $f : X \rightarrow BGL_1(R)$ and a ring map $g : R \rightarrow R'$, then we can extend scalars to obtain that

$$M(BGL_1(g) \circ f) \simeq Mf \otimes_R R'.$$

Consider $\mathbb{S}_p^\wedge \rightarrow \mathbb{F}_p$ which is reduction mod p on homotopy groups. Let h be the induced map under $BGL_1(-)$. Then

$$M(h \circ f_p) \simeq Mf_p \otimes_{\mathbb{S}_p^\wedge} \mathbb{F}_p \simeq Mf_p \otimes \mathbb{F}_p,$$

where the last equivalence comes from the fact that Mf_p and $\mathbb{F}_p$ are p -complete. Explicitly, by the definition of the relative tensor product in $\mathrm{Mod}_{\mathbb{S}_p^\wedge}$, we have that

$$\begin{aligned} Mf_p \otimes_{\mathbb{S}_p^\wedge} \mathbb{F}_p &:= \mathrm{colim}(\cdots \rightrightarrows Mf_p \otimes \mathbb{S}_p^\wedge \otimes \mathbb{F}_p \rightrightarrows Mf_p \otimes \mathbb{F}_p) \\ &\simeq Mf_p \otimes \mathbb{F}_p \end{aligned}$$

because e.g. $\mathbb{S}_p^\wedge \otimes \mathbb{F}_p \simeq \mathbb{S} \otimes \mathbb{F}_p \simeq \mathbb{F}_p$, as $\mathbb{S}_p^\wedge$ is a p -completion and must be p -adically equivalent to $\mathbb{S}$.

So, we have that $M(h \circ f_p) \simeq Mf_p \otimes \mathbb{F}_p$, and furthermore, the composition $h \circ f_p$ gives the trivial $\mathbb{F}_p$ line bundle, since $1+p \equiv 1 \pmod{p}$; in other words, $M(h \circ f_p)$ is the trivial bundle, and so we have the $\mathbb{E}_2$ -equivalence

$$Mf_p \otimes \mathbb{F}_p \simeq \mathbb{F}_p[\Omega^2 S^3],$$

and thus we know the $\mathbb{F}_p$ -homology of Mf_p is also generated in degree 1 under the $\mathbb{E}_2$ Dyer-Lashof operations.

Hence, it suffices to show that the induced map φ_* on $\mathbb{F}_p$ homology is a surjection in degree 1 (since they agree in rank by classical calculations).

Since φ is 1-connected (it is an isomorphism on degree 0, and a surjection on degree 1 since $\pi_1(\mathbb{F}_p) = 0$), and 1-connectivity is preserved by tensoring with connective spectra, we have that

$$Mf_p \otimes \mathbb{F}_p \rightarrow \mathbb{F}_p \otimes \mathbb{F}_p$$

must also be 1-connected and in particular, surjective on degree 1.

3.3. Finishing up the topological proof. Recall the following formula for THH of Thom spectra given by Mattie two lectures ago:

Theorem 3.1. Let R be an $\mathbb{E}_\infty$ ring spectrum and X be a connected $\mathbb{E}_2$ -space. Let $f : X \rightarrow BGL_1(R)$ be an $\mathbb{E}_2$ map, and assume that the $\mathbb{E}_2$ R -algebra structure on Mf extends to an $\mathbb{E}_3$ R -algebra structure. Then we have the following equivalence of $\mathbb{E}_1$ R -algebras

$$\mathrm{THH}(Mf/R) \simeq Mf \otimes \mathbb{S}[BX].$$

Now we simply plug in $Mf_p \simeq \mathbb{F}_p$ to obtain the $\mathbb{E}_1$ -equivalence

$$\mathrm{THH}(\mathbb{F}_p) \simeq \mathbb{F}_p[B\Omega^2 S^3] \simeq \mathbb{F}_p[\Omega S^3]$$

as desired. (Note that $\mathrm{THH}(\mathbb{F}_p/\mathbb{S}_p^\wedge) \simeq \mathrm{THH}(\mathbb{F}_p)$ because their p -completions must agree by definition, as $\mathbb{F}_p$ is p -complete, and they are both already p -complete.)

4. A COMPARISON OF THE PROOFS

The ideas of this discussion originate in [KN19, Appendix A.]. In some intuitive sense, both of the arguments we presented are already related in spirit by the fact that they reduce the result to a computation on degree 0, 1 $\mathbb{F}_p$ -homology, since they both apply Steinberger's computations of Dyer-Lashof operations on the dual Steenrod algebra.

Whereas the algebraic argument directly characterizes the dual Steenrod algebra as a free $\mathbb{E}_2$ - $\mathbb{F}_p$ -algebra, the topological argument first shows that $\mathbb{F}_p$ (viewed as an $\mathbb{E}_2$ -ring) arises as a Thom spectrum.

The connection between the two is discussed in [ACB], which characterizes Thom spectra of spheres as “weakly universal” algebras, i.e. so-called versal algebras; basically, $\mathbb{F}_p$ being a Thom spectrum means that it should look like a free object. In particular, they define:

Definition 4.1. Let R be an $\mathbb{E}_{n+1}$ ring spectrum. Given $\chi : \Sigma^k R \rightarrow R$, the *versal R -algebra $R //_{\mathbb{E}_n} \chi$ of characteristic χ* is the following pushout in $\mathrm{Alg}_R^{\mathbb{E}_n}$:

$$\begin{array}{ccc} \mathrm{Free}_{\mathbb{E}_n}(\Sigma^k R) & \xrightarrow{\tilde{\theta}} & R \\ \tilde{\chi} \downarrow & \ulcorner & \downarrow \\ R & \longrightarrow & R //_{\mathbb{E}_n} \chi \end{array}$$

where $\mathrm{Free}_{\mathbb{E}_n}$ denotes the free $\mathbb{E}_n$ -algebra functor $\mathrm{Mod}_R \rightarrow \mathrm{Alg}_R^{\mathbb{E}_n}$ which is left adjoint to the forgetful functor, and the tildes denote adjoint maps.

A map $f : S^{k+1} \rightarrow BGL_1 R$ induces a map $\tilde{f} : \Sigma^k R \rightarrow R$ in the following way: first we obtain a map

$$S^k \rightarrow GL_1(R) \hookrightarrow \Omega^\infty R$$

by suspension-loop adjunction, bar-loop equivalence, and inclusion into Ω^∞ . Then after suspending, we obtain the spectra map $\Sigma^k \mathbb{S} \rightarrow R$, and this gives us an R -linear map $\Sigma^k R \rightarrow R$. Now we define $\chi(f) : \Sigma^k R \rightarrow R$, the characteristic associated to f , as follows:

$$\chi(f) = \begin{cases} \tilde{f} & \text{if } k > 0 \\ \tilde{f} - 1 & \text{if } k = 0. \end{cases}$$

One can take the Thom spectrum of the corresponding n -fold loop map to obtain an $\mathbb{E}_n$ -algebra, or one can take the versal algebra. In particular, it turns out that we have the following result [ACB, Theorem 4.10]:

Theorem 4.1. Given a map $S^{k+1} \rightarrow BGL_1 R$ with corresponding characteristic χ and corresponding n -fold loop map $f : \Omega^n \Sigma^n S^{k+1} \rightarrow BGL_1 R$, we have the following equivalence of $\mathbb{E}_n$ R -algebras:

$$Mf \simeq R //_{\mathbb{E}_n} \chi.$$

This is proven by demonstrating that both spectra corepresent the same functor on $\mathrm{Alg}_R^{\mathbb{E}_n}$: it turns out that for all $A \in \mathrm{Alg}_R^{\mathbb{E}_n}$ with characteristic χ , i.e. algebras such that composition with the unit $\Sigma^k R \rightarrow R \rightarrow A$ is nullhomotopic (the mapping spaces are empty otherwise),

$$\mathrm{Map}_{\mathrm{Alg}_R^{\mathbb{E}_n}}(Mf, A) \simeq \mathrm{Map}_{\mathrm{Alg}_R^{\mathbb{E}_n}}(R //_{\mathbb{E}_n} \chi, A) \simeq \Omega^{\infty+k+1} A,$$

which uses the fact that the $\mathbb{E}_n$ structure of Thom spectra has a universal characterization [ACB, Theorem 3.5].

In our case, for our map $f_p : \Omega^2 S^3 \rightarrow BGL_1(\mathbb{S}_p^\wedge)$, we have that

$$Mf_p \simeq \mathbb{S}_p^\wedge //_{\mathbb{E}_2} p$$

where we denote the characteristic of f_p by p , since it is given by $p+1-1 = p$ as a map $\mathbb{S}_p^\wedge \rightarrow \mathbb{S}_p^\wedge$. Furthermore, the pushout and free construction both commute with $- \otimes \mathbb{F}_p$, and since $\mathbb{S}_p^\wedge \otimes \mathbb{F}_p \simeq \mathbb{F}_p$, we obtain the pushout on the right:

$$\begin{array}{ccc} \mathrm{Free}_{\mathbb{E}_2}(\mathbb{S}_p^\wedge) & \xrightarrow{0} & \mathbb{S}_p^\wedge \\ p \downarrow & \lrcorner & \downarrow \\ \mathbb{S}_p^\wedge & \longrightarrow & \mathbb{S}_p^\wedge //_{\mathbb{E}_2} p \end{array} \quad \rightsquigarrow \quad \begin{array}{ccc} \mathrm{Free}_{\mathbb{E}_2}(\mathbb{F}_p) & \xrightarrow{0} & \mathbb{F}_p \\ 0 \downarrow & \lrcorner & \downarrow \\ \mathbb{F}_p & \longrightarrow & Mf_p \otimes \mathbb{F}_p \end{array}.$$

Simultaneously, it is true that we have the pushout

$$\begin{array}{ccc} \mathrm{Free}_{\mathbb{E}_2}(\mathbb{F}_p) & \xrightarrow{0} & \mathbb{F}_p \\ 0 \downarrow & \lrcorner & \downarrow \\ \mathbb{F}_p & \longrightarrow & \mathrm{Free}_{\mathbb{E}_2}(\Sigma\mathbb{F}_p) \end{array}$$

by definition of suspension as a pushout. Hence, showing $Mf_p \otimes \mathbb{F}_p \simeq \mathbb{F}_p \otimes \mathbb{F}_p$ is exactly equivalent to showing that

$$\mathrm{Free}_{\mathbb{E}_2}(\Sigma\mathbb{F}_p) \simeq \mathbb{F}_p \otimes \mathbb{F}_p$$

where the left-hand side is exactly $\mathbb{F}_p[\Omega^2 S^3]$, i.e. the free $\mathbb{E}_2$ $\mathbb{F}_p$ -algebra on a generator in degree 1. Explicitly, we see that:

$$\begin{aligned} \mathrm{Map}_{\mathrm{Alg}_{\mathbb{F}_p}^{\mathbb{E}_2}}(\mathrm{Free}_{\mathbb{E}_2}(\Sigma\mathbb{F}_p), A) &\simeq \mathrm{Map}_{\mathrm{Mod}_{\mathbb{F}_p}}(\Sigma\mathbb{F}_p, A) \\ &\simeq \mathrm{Map}_{\mathrm{Mod}_{\mathbb{F}_p}}(\Sigma\mathbb{S} \otimes \mathbb{F}_p, A) \\ &\simeq \mathrm{Map}_{\mathrm{Sp}}(\Sigma\mathbb{S}, A) \\ &\simeq \mathrm{Map}_{\mathcal{S}_*}(S^1, \Omega^\infty A) \\ &\simeq \mathrm{Map}_{\mathrm{Alg}_{\mathbb{F}_p}^{\mathbb{E}_2}}(\mathbb{F}_p[\Omega^2 S^3], A). \end{aligned}$$

Hence, the proof of Bökstedt using the Hopkins-Mahowald result is equivalent to the algebraic argument. In other words, since the Thom spectra of spheres are characterized by their multiplicativity in a “versal” way, this can sometimes allow us to simplify the “topological” arguments that use them into arguments that feel more “purely algebraic”.

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